Shooting Lab EDA

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Data Set:

```r
library(readr)

shooting_data <- read_csv("capstone2026.csv")
```

```
## Rows: 25868 Columns: 262
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr   (5): Name, Date, Shot.Type, Shot.Location, PlayerID
## dbl (256): Shot.Distance, Shot.X.Pos, Shot.Y.Pos, Ball_Depth, left_right, Ba...
## lgl   (1): Made
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
shooting_data %>%
  group_by(Name) %>%
  count()
```

```
## # A tibble: 165 x 2
## # Groups:   Name [165]
##    Name          n
##    <chr>     <int>
##  1 Player 1     77
```

```
##   2 Player 10    130
##   3 Player 100     3
##   4 Player 101   106
##   5 Player 102  1494
##   6 Player 103   326
##   7 Player 104   150
##   8 Player 105    55
##   9 Player 106    59
##  10 Player 107    47
## # i 155 more rows
```

Filter for Freethrows and more than 20 shots:

```r
freethrows = shooting_data %>%
  filter(Shot.Location == "Free Throw")

head(freethrows)
```

```
## # A tibble: 6 x 262
##   Name   Date  Shot.Type Shot.Location Shot.Distance Shot.X.Pos Shot.Y.Pos Made
##   <chr>  <chr> <chr>     <chr>                 <dbl>      <dbl>      <dbl> <lgl>
## 1 Playe~ 5/27~ Off the ~ Free Throw             13.8      0.750       13.8 TRUE
## 2 Playe~ 5/27~ Off the ~ Free Throw             13.8      0.849       13.8 TRUE
## 3 Playe~ 5/27~ Off the ~ Free Throw             13.8      0.714       13.8 TRUE
## 4 Playe~ 5/27~ Off the ~ Free Throw             13.9      0.736       13.8 TRUE
## 5 Playe~ 5/27~ Off the ~ Free Throw             13.8      0.707       13.8 TRUE
## 6 Playe~ 5/27~ Off the ~ Free Throw             13.8      0.800       13.8 TRUE
## # i 254 more variables: Ball_Depth <dbl>, left_right <dbl>,
## #   Ball.Distance.from.Center <dbl>, Ball_to_Rim_Angle <dbl>,
## #   Ball_Velocity <dbl>, Release.Height <dbl>, Initial.Ball.Angle <dbl>,
## #   Initial.Ball.Velocity <dbl>, Starting.Ball.Height <dbl>,
## #   Max.Ball.Arc <dbl>, Pass.Direction <dbl>, timeAtStartofMovement <dbl>,
## #   timeAtStartofHitch <dbl>, timeAtEndofHitch <dbl>, timeAtRelease <dbl>,
## #   TotalShotTime <dbl>, Pre.Hitch.Time <dbl>, Hitch.Time <dbl>, ...
```

```r
summary(freethrows$Made)
```

```
##    Mode   FALSE    TRUE
## logical     716    3251
```

```r
freethrows %>%
  group_by(PlayerID) %>%
  count()
```

```
## # A tibble: 136 x 2
## # Groups:   PlayerID [136]
##    PlayerID       n
##    <chr>      <int>
##  1 Player 1      19
##  2 Player 10     32
```

```
##   3 Player 100     2
##   4 Player 101    26
##   5 Player 102   100
##   6 Player 103    52
##   7 Player 104    32
##   8 Player 106     6
##   9 Player 107    17
##  10 Player 108     5
## # i 126 more rows
```

```
ft_counts <- shooting_data %>%
  filter(Shot.Location == "Free Throw") %>%
  count(Name, name = "n_free_throws") %>%
  arrange(desc(n_free_throws))

ft_counts
```

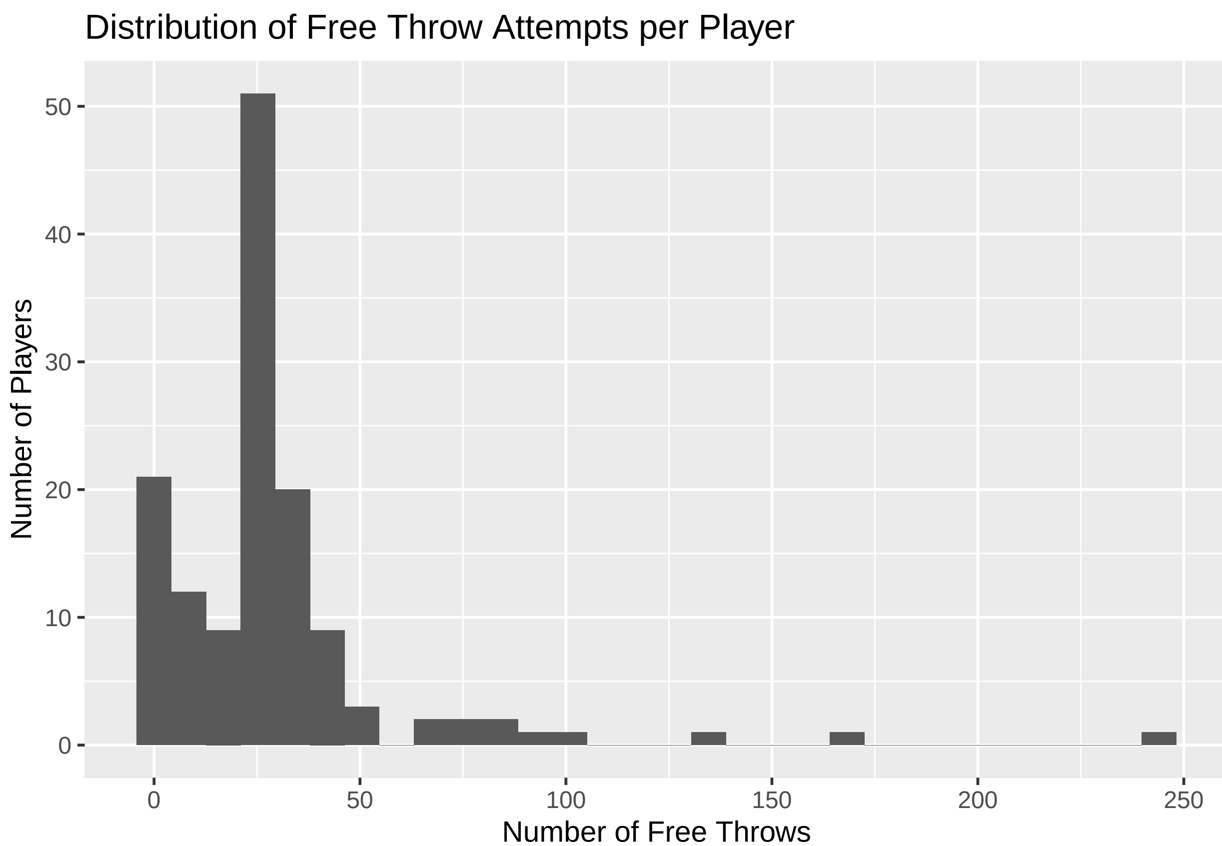
```
## # A tibble: 136 x 2
##    Name       n_free_throws
##    <chr>              <int>
##  1 Player 116           245
##  2 Player 163           172
##  3 Player 158           135
##  4 Player 102           100
##  5 Player 120            93
##  6 Player 145            80
##  7 Player 64             80
##  8 Player 54             78
##  9 Player 23             74
## 10 Player 98             66
## # i 126 more rows
```

```
freethrows_20 <- shooting_data %>%
  filter(Shot.Location == "Free Throw") %>%
  group_by(Name) %>%
  filter(n() >= 20) %>%
  ungroup()

freethrows_20 <- freethrows_20 %>%
  mutate(
    made_num = ifelse(Made == TRUE, 1, 0)
  )
```

Distribution of Shots

```
library(ggplot2)

shooting_data %>%
  filter(Shot.Location == "Free Throw") %>%
```

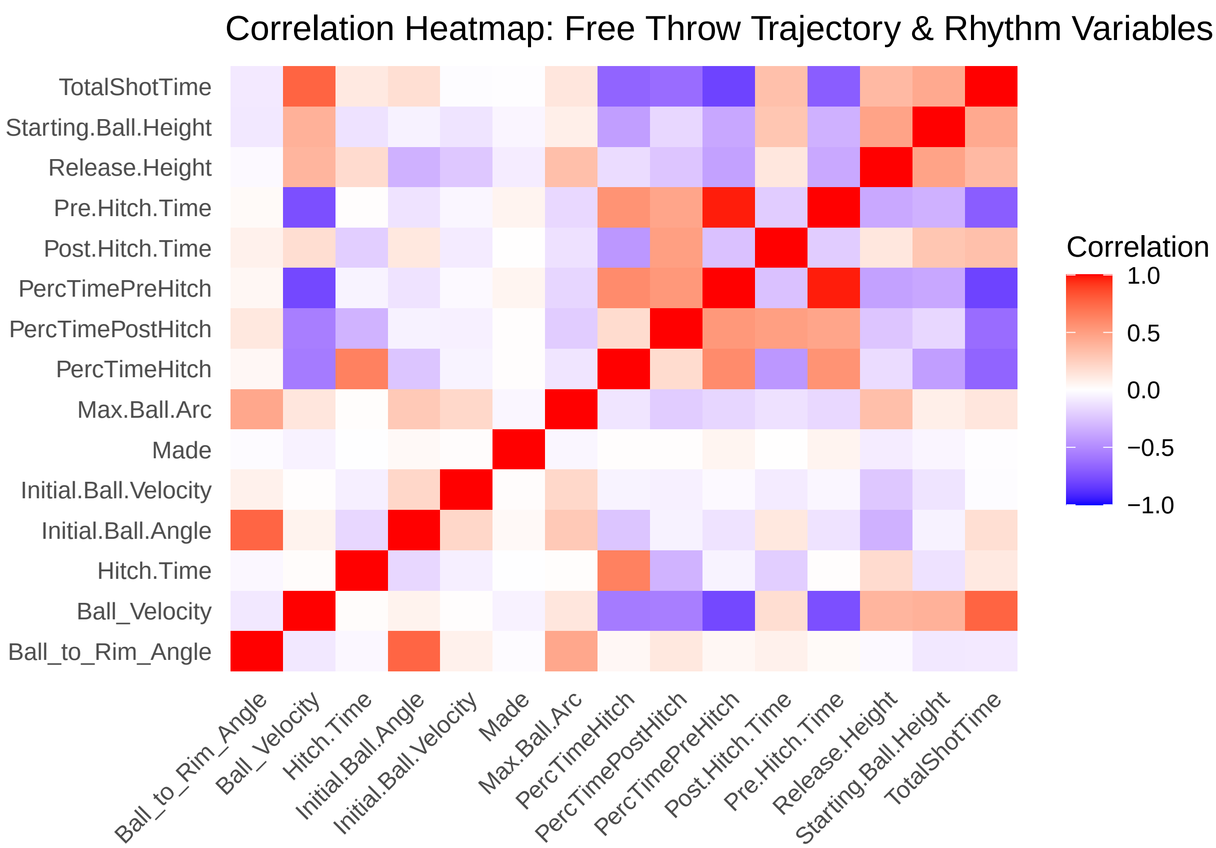
```
  count(Name, name = "n_shots") %>%
  ggplot(aes(x = n_shots)) +
  geom_histogram(bins = 30) +
  labs(
    title = "Distribution of Free Throw Attempts per Player",
    x = "Number of Free Throws",
    y = "Number of Players"
  )
```



Variable Correlations

```
library(tidyr)

shot_corrs <- freethrows_20 %>%
  select(
    Made,
    Ball_to_Rim_Angle,
    Ball_Velocity,
    Initial.Ball.Angle,
    Initial.Ball.Velocity,
    Starting.Ball.Height,
    Release.Height,
    Max.Ball.Arc,
```

```r
    TotalShotTime,
    Pre.Hitch.Time,
    Hitch.Time,
    Post.Hitch.Time,
    PercTimePreHitch,
    PercTimeHitch,
    PercTimePostHitch
  )

cor_matrix <- cor(shot_corrs, use = "complete.obs")

cor_long <- as.data.frame(cor_matrix) %>%
  mutate(Variable1 = rownames(.)) %>%
  pivot_longer(
    cols = -Variable1,
    names_to = "Variable2",
    values_to = "Correlation"
  )

ggplot(cor_long, aes(x = Variable1, y = Variable2, fill = Correlation)) +
  geom_tile() +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0,
    limits = c(-1, 1)
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.grid = element_blank()
  ) +
  labs(
    title = "Correlation Heatmap: Free Throw Trajectory & Rhythm Variables",
    x = NULL,
    y = NULL
  )
```




```r
library(tidyverse)
```

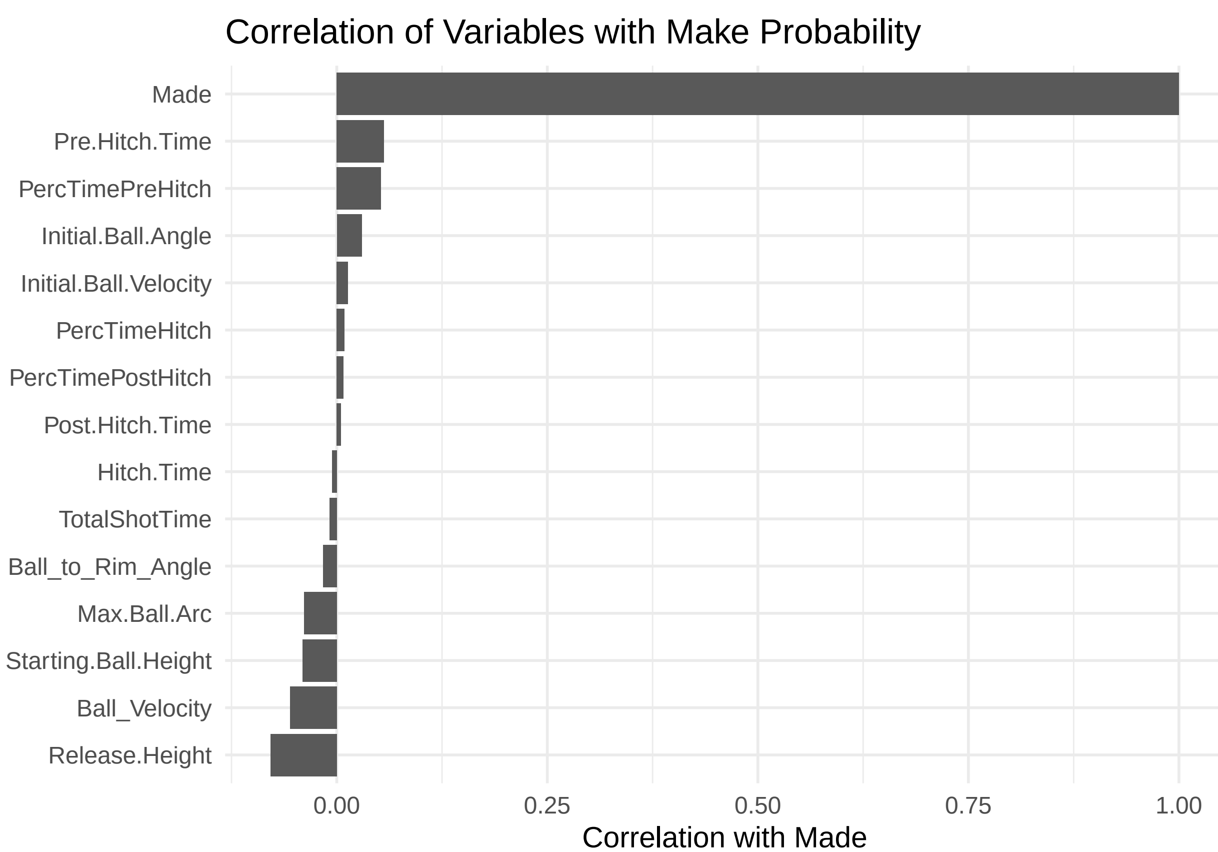
```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v stringr   1.5.1
## v lubridate 1.9.3     v tibble    3.2.1
## v purrr     1.1.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
made_corr <- cor_matrix[, "Made"] %>%
  as.data.frame() %>%
  rownames_to_column("Variable")

colnames(made_corr)[2] <- "Correlation_with_Made"

made_corr %>%
  arrange(desc(abs(Correlation_with_Made)))
```

```
##                 Variable Correlation_with_Made
## 1                   Made           1.000000000
## 2         Release.Height          -0.078623419
## 3          Pre.Hitch.Time           0.056086938
```

```
## 4          Ball_Velocity          -0.055660258
## 5        PercTimePreHitch          0.052678164
## 6    Starting.Ball.Height          -0.040579798
## 7            Max.Ball.Arc          -0.038977859
## 8       Initial.Ball.Angle          0.029616762
## 9       Ball_to_Rim_Angle          -0.016148634
## 10 Initial.Ball.Velocity          0.013197021
## 11          PercTimeHitch          0.009291658
## 12          TotalShotTime          -0.008499761
## 13       PercTimePostHitch          0.008134937
## 14             Hitch.Time          -0.005468104
## 15        Post.Hitch.Time          0.005087644
```

```
ggplot(made_corr, aes(
  x = reorder(Variable, Correlation_with_Made),
  y = Correlation_with_Made
)) +
  geom_col() +
  coord_flip() +
  theme_minimal() +
  labs(
    title = "Correlation of Variables with Make Probability",
    x = NULL,
    y = "Correlation with Made"
  )
```



```r
var_summary = freethrows_20 %>%
  group_by(Made) %>%
  summarise(
    across(
      c(Ball_to_Rim_Angle,
        Ball_Velocity,
        Initial.Ball.Angle,
        Initial.Ball.Velocity,
        Starting.Ball.Height,
        Release.Height,
        Max.Ball.Arc,
        TotalShotTime,
        Pre.Hitch.Time,
        Hitch.Time,
        Post.Hitch.Time,
        PercTimePreHitch,
        PercTimeHitch,
        PercTimePostHitch),
      list(
        mean = ~mean(.x, na.rm = TRUE),
        sd   = ~sd(.x, na.rm = TRUE)
      )
    ),
    n = n()
  )

var_summary
```

```
## # A tibble: 2 x 30
##   Made  Ball_to_Rim_Angle_mean Ball_to_Rim_Angle_sd Ball_Velocity_mean
##   <lgl>                  <dbl>                <dbl>              <dbl>
## 1 FALSE                   45.0                 3.39               3.40
## 2 TRUE                    44.9                 3.00               3.20
## # i 26 more variables: Ball_Velocity_sd <dbl>, Initial.Ball.Angle_mean <dbl>,
## #   Initial.Ball.Angle_sd <dbl>, Initial.Ball.Velocity_mean <dbl>,
## #   Initial.Ball.Velocity_sd <dbl>, Starting.Ball.Height_mean <dbl>,
## #   Starting.Ball.Height_sd <dbl>, Release.Height_mean <dbl>,
## #   Release.Height_sd <dbl>, Max.Ball.Arc_mean <dbl>, Max.Ball.Arc_sd <dbl>,
## #   TotalShotTime_mean <dbl>, TotalShotTime_sd <dbl>,
## #   Pre.Hitch.Time_mean <dbl>, Pre.Hitch.Time_sd <dbl>, ...
```

```r
library(dplyr)
library(tidyr)

# Separate made and miss rows
made_stats  <- var_summary %>% filter(Made == TRUE)
miss_stats  <- var_summary %>% filter(Made == FALSE)

# Extract means
made_means <- made_stats %>%
  select(ends_with("_mean")) %>%
  pivot_longer(everything(),
               names_to = "variable",
```

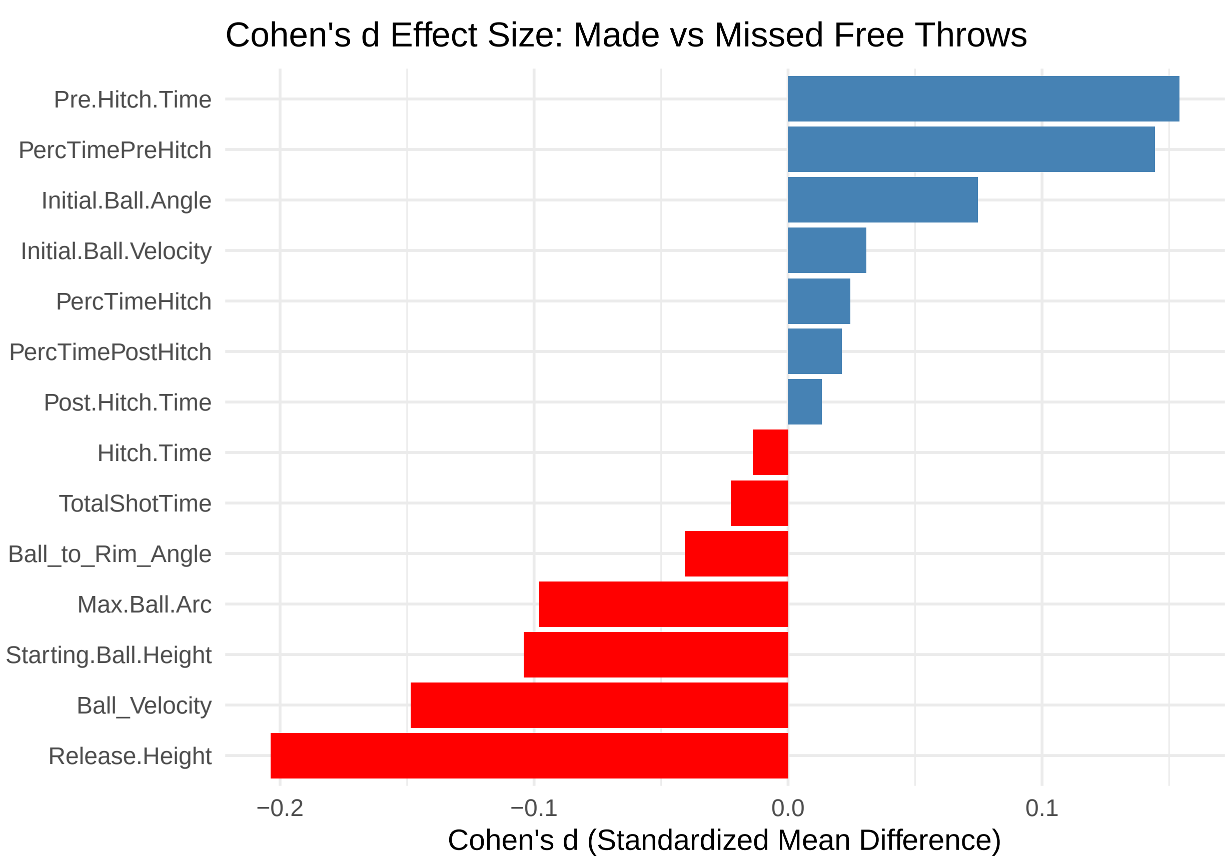
```r
               values_to = "mean_made") %>%
  mutate(variable = sub("_mean", "", variable))

miss_means <- miss_stats %>%
  select(ends_with("_mean")) %>%
  pivot_longer(everything(),
               names_to = "variable",
               values_to = "mean_miss") %>%
  mutate(variable = sub("_mean", "", variable))

# Extract sds
made_sd <- made_stats %>%
  select(ends_with("_sd")) %>%
  pivot_longer(everything(),
               names_to = "variable",
               values_to = "sd_made") %>%
  mutate(variable = sub("_sd", "", variable))

miss_sd <- miss_stats %>%
  select(ends_with("_sd")) %>%
  pivot_longer(everything(),
               names_to = "variable",
               values_to = "sd_miss") %>%
  mutate(variable = sub("_sd", "", variable))

# Combine
effect_sizes <- made_means %>%
  left_join(miss_means, by = "variable") %>%
  left_join(made_sd, by = "variable") %>%
  left_join(miss_sd, by = "variable") %>%
  mutate(
    pooled_sd = sqrt((sd_made^2 + sd_miss^2)/2),
    cohens_d = (mean_made - mean_miss)/pooled_sd,
    abs_d = abs(cohens_d)
  ) %>%
  arrange(desc(abs_d))

effect_sizes
```

```
## # A tibble: 14 x 8
##    variable        mean_made mean_miss sd_made sd_miss pooled_sd cohens_d  abs_d
##    <chr>               <dbl>     <dbl>   <dbl>   <dbl>     <dbl>    <dbl>  <dbl>
##  1 Release.Height      2.54      2.56    0.128   0.134     0.131  -0.204  0.204
##  2 Pre.Hitch.Time      0.641     0.555   0.595   0.514     0.556   0.154  0.154
##  3 Ball_Velocity       3.20      3.40    1.38    1.30      1.34   -0.149  0.149
##  4 PercTimePreHit~     1.19      1.00    1.36    1.18      1.27    0.144  0.144
##  5 Starting.Ball.~     1.08      1.10    0.211   0.227     0.219  -0.104  0.104
##  6 Max.Ball.Arc        3.94      3.96    0.215   0.244     0.230  -0.0980 0.0980
##  7 Initial.Ball.A~    48.5      48.2     2.94    3.29      3.12    0.0748 0.0748
##  8 Ball_to_Rim_An~    44.9      45.0     3.00    3.39      3.20   -0.0406 0.0406
##  9 Initial.Ball.V~     6.55      6.54    0.214   0.298     0.259   0.0308 0.0308
## 10 PercTimeHitch       0.377     0.374   0.139   0.136     0.138   0.0245 0.0245
## 11 TotalShotTime       0.696     0.700   0.167   0.163     0.165  -0.0224 0.0224
```

```
## 12 PercTimePostHi~      0.373      0.370  0.113   0.116      0.115    0.0212 0.0212
## 13 Hitch.Time           0.246      0.247  0.0707  0.0778     0.0743  -0.0139 0.0139
## 14 Post.Hitch.Time      0.247      0.247  0.0628  0.0647     0.0637   0.0132 0.0132
```

Standardized mean differences between made and missed free throws were uniformly small ($|d| < 0.21$), indicating that no single trajectory or timing variable independently explains shot success. This suggests that shot outcome is likely influenced by multivariate interactions rather than dominant individual predictors.

```r
library(ggplot2)

ggplot(effect_sizes, aes(
  x = reorder(variable, cohens_d),
  y = cohens_d,
  fill = cohens_d > 0
)) +
  geom_col() +
  coord_flip() +
  scale_fill_manual(values = c("red", "steelblue"), guide = "none") +
  theme_minimal() +
  labs(
    title = "Cohen's d Effect Size: Made vs Missed Free Throws",
    x = NULL,
    y = "Cohen's d (Standardized Mean Difference)"
  )
```



Pre.Hitch.Time and PercTimePreHitch:

- Slightly higher for made shots
- makes tend to have a slightly longer setup phase.

Release.Height and Ball_Velocity

- Slightly higher for missed shots
- misses may be released a bit higher and enter the rim slightly faster.

```r
player_summary <- freethrows_20 %>%
  group_by(Name) %>%
  summarise(
    attempts = n(),
    makes = sum(made_num),
    FT_pct = mean(Made),
    avg_entry_angle = mean(Ball_to_Rim_Angle),
    avg_arc = mean(Max.Ball.Arc),
    avg_release_angle = mean(Initial.Ball.Angle),
    avg_velocity = mean(Initial.Ball.Velocity),
    avg_total_time = mean(TotalShotTime),
    .groups = "drop"
  ) %>%
  arrange(desc(FT_pct))

player_summary
```

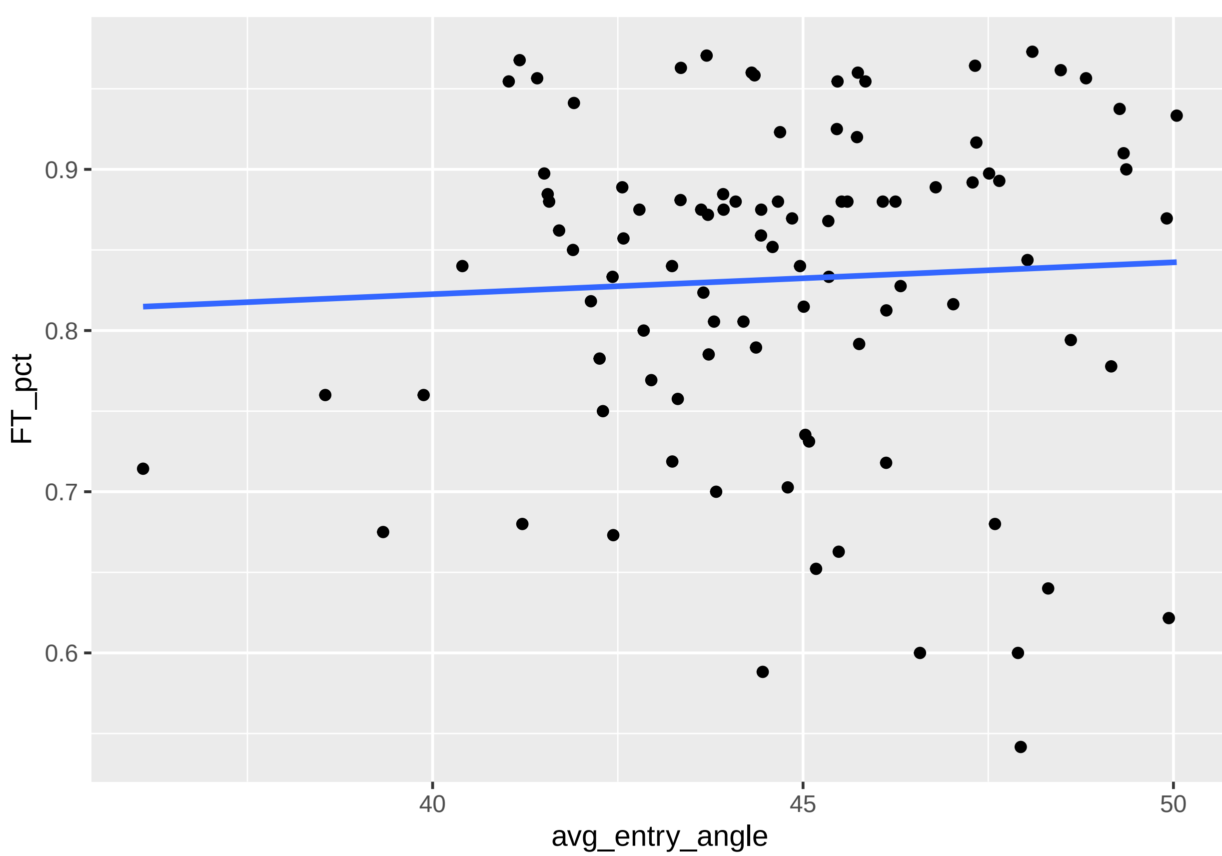
```
## # A tibble: 97 x 9
##    Name       attempts makes FT_pct avg_entry_angle avg_arc avg_release_angle
##    <chr>         <int> <dbl>  <dbl>           <dbl>   <dbl>             <dbl>
##  1 Player 68        37    36  0.973            48.1    4.16              52.8
##  2 Player 20        34    33  0.971            43.7    3.95              48.6
##  3 Player 147       31    30  0.968            41.2    3.83              47.2
##  4 Player 162       28    27  0.964            47.3    3.95              47.5
##  5 Player 73        27    26  0.963            43.4    3.99              48.0
##  6 Player 101       26    25  0.962            48.5    4.24              51.3
##  7 Player 14        25    24  0.96             45.7    3.90              47.4
##  8 Player 160       25    24  0.96             44.3    3.76              48.9
##  9 Player 18        24    23  0.958            44.3    3.85              48.3
## 10 Player 110       23    22  0.957            41.4    3.74              43.9
## # i 87 more rows
## # i 2 more variables: avg_velocity <dbl>, avg_total_time <dbl>
```

```r
library(knitr)
player_summary_corr <- player_summary %>%
  select(-Name) %>%
  cor(use = "complete.obs")

kable(player_summary_corr)
```

	attempts	makes	FT_pct	avg_entry_angle	avg_arc	avg_release_angle	avg_velocity	avg_total_time
attempts	1.0000000	0.9861001	-0.1014963	0.0806538	0.0454343	0.0920546	0.1433251	0.2028130
makes	0.9861001	1.0000000	0.0350878	0.0936059	0.0324079	0.1284937	0.1719497	0.2017040
FT_pct	-0.1014963	0.0350878	1.0000000	0.0524340	-0.0479984	0.1764490	0.1798612	-0.0883755
avg_entry_angle	0.0806538	0.0936059	0.0524340	1.0000000	0.5649433	0.7733363	0.1803424	-0.0682254
avg_arc	0.0454343	0.0324079	-0.0479984	0.5649433	1.0000000	0.4525391	0.1073298	0.3180805
avg_release_angle	0.0920546	0.1284937	0.1764490	0.7733363	0.4525391	1.0000000	0.4802947	0.1672820
avg_velocity	0.1433251	0.1719497	0.1798612	0.1803424	0.1073298	0.4802947	1.0000000	-0.0298858
avg_total_time	0.2028130	0.2017040	-0.0883755	-0.0682254	0.3180805	0.1672820	-0.0298858	1.0000000

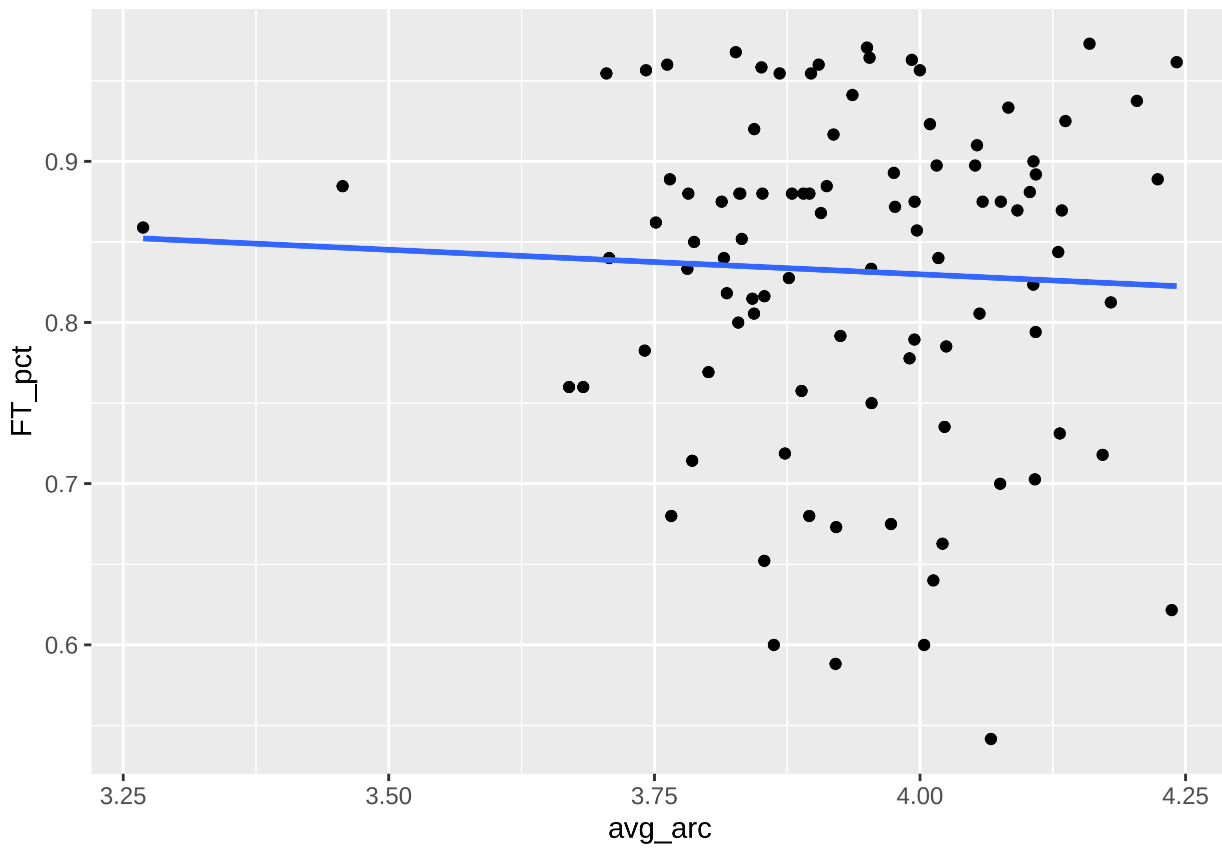
```r
ggplot(player_summary,
       aes(x = avg_entry_angle, y = FT_pct)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



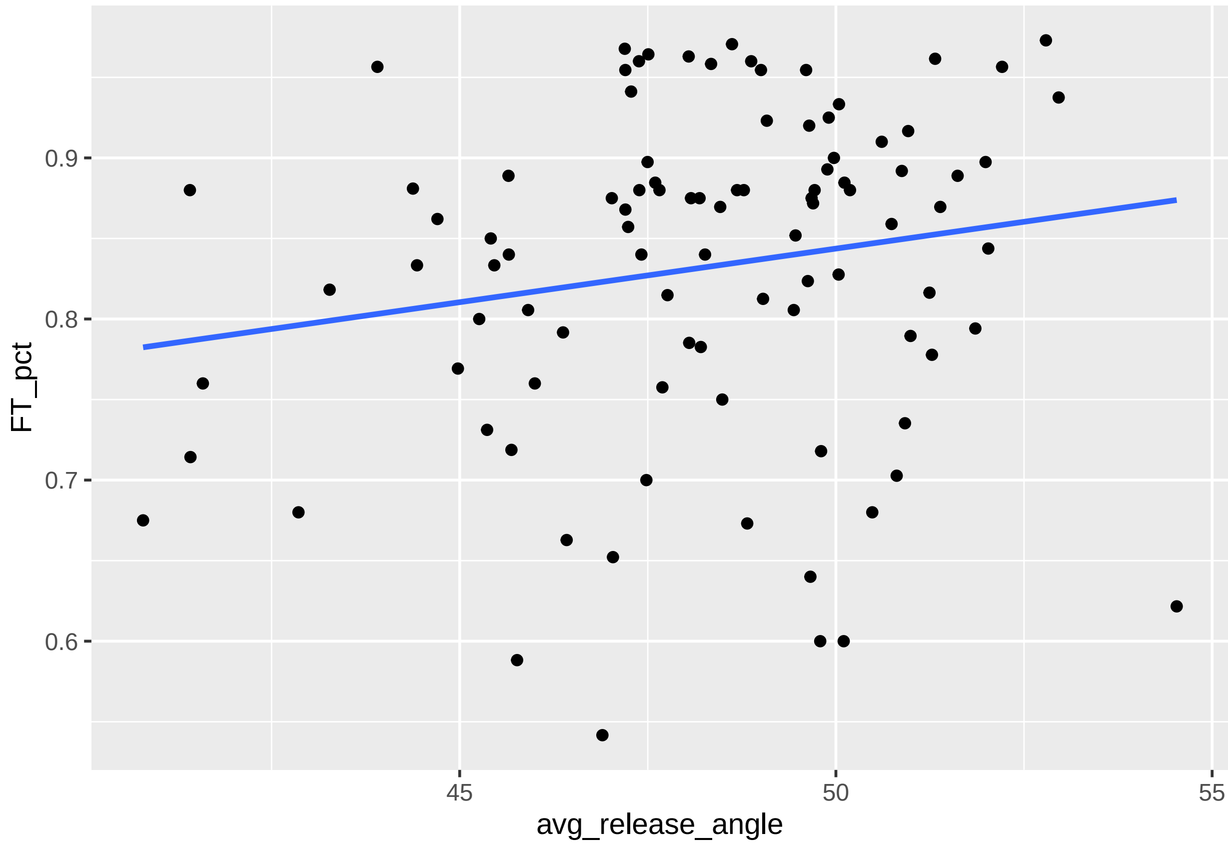
```r
ggplot(player_summary,
       aes(x = avg_arc, y = FT_pct)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



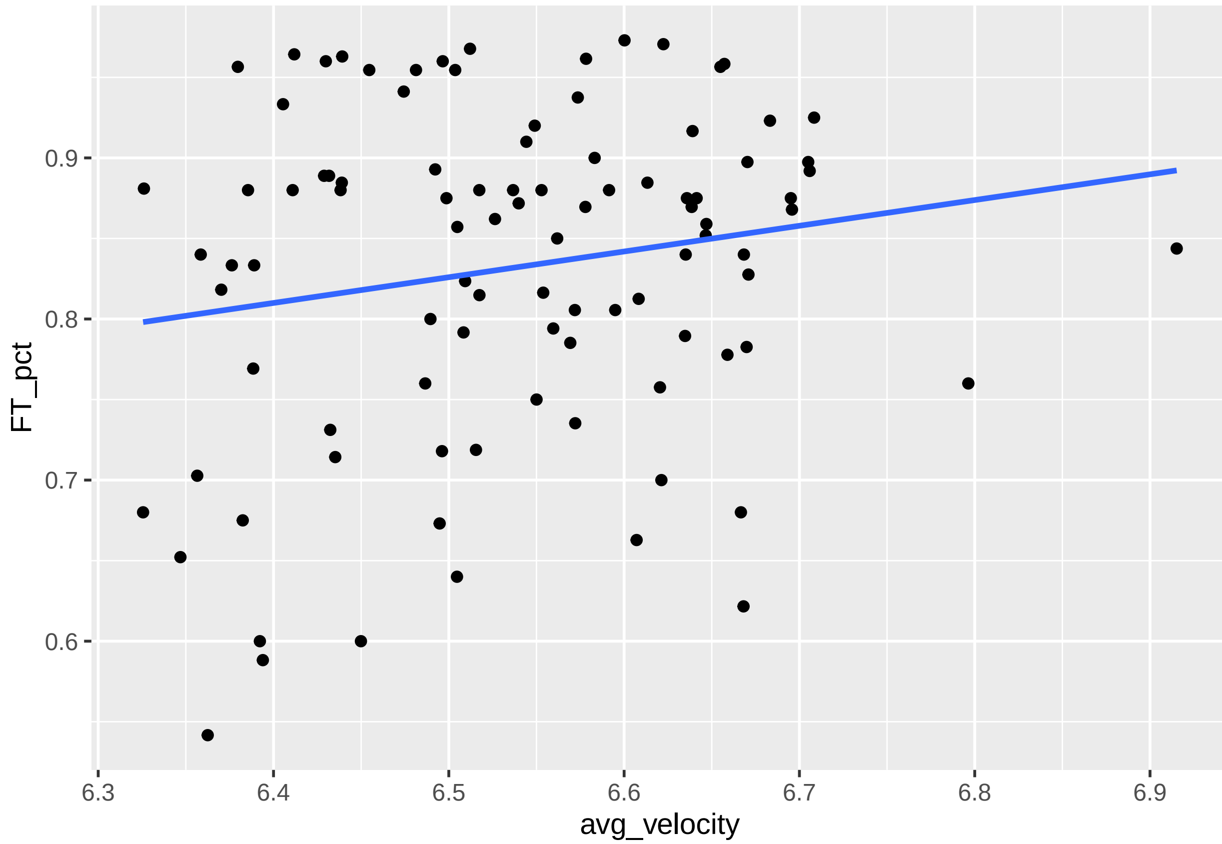

```r
ggplot(player_summary,
       aes(x = avg_release_angle    , y = FT_pct)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



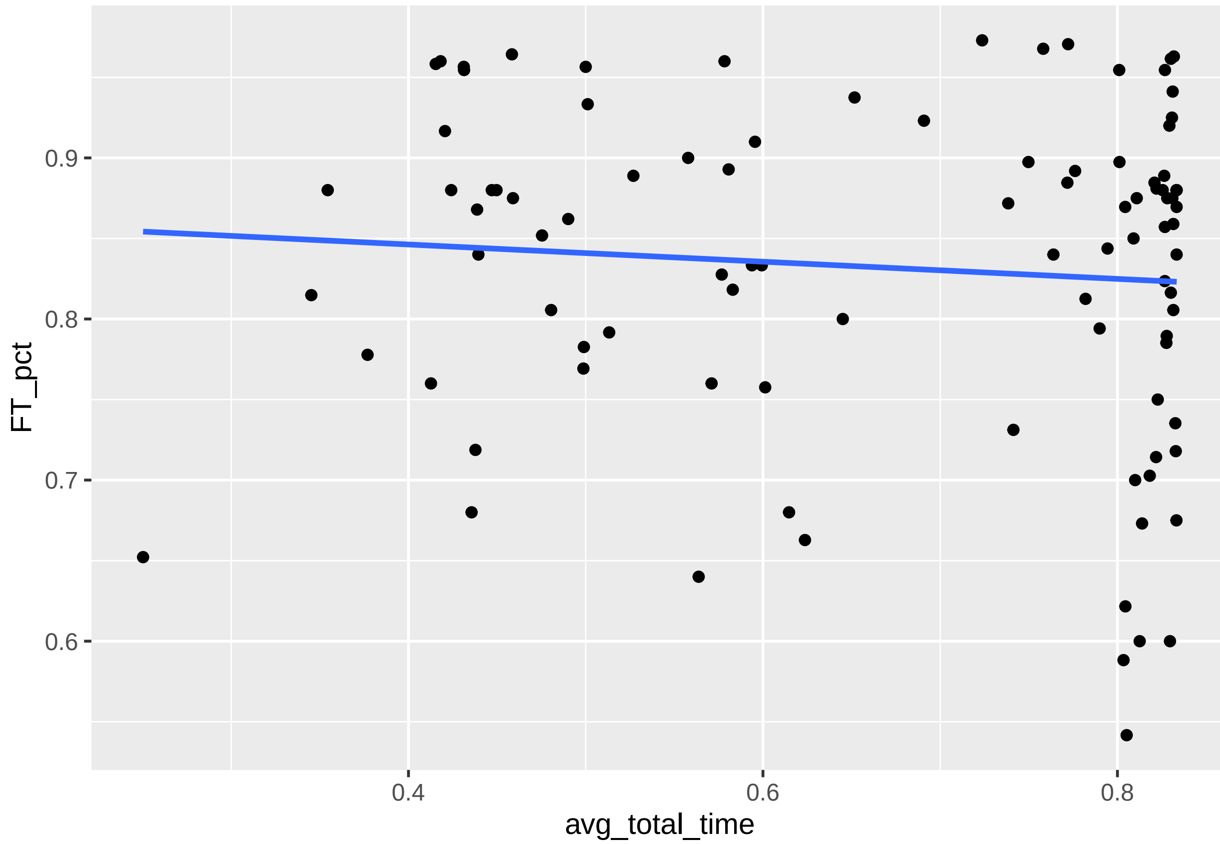

```
ggplot(player_summary,
       aes(x = avg_velocity, y = FT_pct)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```




```
ggplot(player_summary,
       aes(x = avg_total_time, y = FT_pct)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



add other vars

```
freethrows_20 %>%
  group_by(Shot.Type) %>%
  count()
```

```
## # A tibble: 2 x 2
## # Groups:   Shot.Type [2]
##   Shot.Type            n
##   <chr>            <int>
## 1 Catch and Shoot    452
## 2 Off the Dribble   3264
```